Research Proposal

Project Title: Multimodal Machine Learning for Heart Failure Prediction and Survival Analysis

Domain / Category: Healthcare AI / Clinical Machine Learning

Author Name: Pranav Ananth

Email: pranav_ananth@hotmail.com

With the domain and research question selected, this proposal details the finalized research idea to receive feedback from the mentor and program team. The sections below outline the problem, anticipated challenges, datasets, technical approaches, and evaluation strategies for this project.

1. Research Problem

Heart failure remains one of the leading causes of mortality worldwide, yet current clinical prediction models often rely on limited, structured data and fail to capture the full complexity of a patient's medical profile. This project aims to develop a robust, multimodal machine learning system for better prediction of heart failure outcomes and patient survival by integrating a wide range of data sources — including structured clinical laboratory results, body measurements, wearable device data from tools like the Apple Watch, past medical records from EHR systems such as EPIC, and unstructured physician notes.

Why this project: Heart failure prediction is a high-stakes problem where marginal improvements in model accuracy can directly translate to lives saved. The integration of unstructured and multimodal data sources opens a compelling frontier that goes well beyond what most existing models attempt.

Why it is interesting: The intersection of graph-based deep learning, natural language processing for clinical notes, and wearable sensor data presents a rich, technically challenging problem. It also has the potential to democratize high-quality cardiac risk assessment by incorporating data streams that are already widely collected but underutilized.

2. Anticipated Challenges

Several key challenges are expected throughout this project:

- Small sample sizes: Heart failure datasets with comprehensive multimodal records are limited in volume, which may hinder model generalization and increase the risk of overfitting.
- High-dimensional features: Combining clinical lab results, wearable signals, imaging, and text-derived features creates a high-dimensional input space, requiring careful dimensionality reduction and feature selection strategies.

- Acquisition of physician notes: Unstructured doctor notes are sensitive, often proprietary, and may not be readily accessible due to privacy regulations (e.g., HIPAA). De-identification and IRB approvals may be required.
- ECG and historical EHR data: Integrating time-series ECG readings and longitudinal patient histories from EPIC or similar systems requires standardization across disparate formats (HL7, FHIR, proprietary), which adds significant preprocessing complexity.

3. Dataset Description

This project will draw on the following datasets and data sources:

- Heart Failure Clinical Records Dataset: A publicly available dataset containing 299 patient records with 13 clinical features (e.g., serum creatinine, ejection fraction, age, anaemia, diabetes) and a death event target variable. Originally used by Chicco and Jurman (2020), it is available through the UCI Machine Learning Repository.
- EPIC EHR Data (Past Patient History): Longitudinal patient records from EPIC, including diagnoses, medications, lab results, and visit histories. Access will be sought through academic medical center partnerships or de-identified data sharing agreements.
- Unstructured Doctor Notes: Clinical free-text notes from EHR systems. These will be collected through approved data-sharing arrangements with hospital partners, subject to de-identification protocols consistent with HIPAA guidelines.
- Wearable Device Data (Apple Watch): Heart rate time-series, activity levels, resting pulse, and irregular rhythm notifications. Where real patient data is unavailable, publicly available wearable health datasets (e.g., MIMIC-IV Waveform, PhysioNet) will serve as proxies.
- ECG Data: 12-lead electrocardiogram recordings will be sourced from publicly available repositories including the PTB-XL ECG Dataset (PhysioNet), which contains over 21,000 clinical 12-lead ECGs with physician-annotated labels. Additional ECG time-series may be drawn from the MIMIC-IV Waveform Database. These recordings will be used to extract features related to cardiac rhythm abnormalities, QRS morphology, ST-segment changes, and heart rate variability — all clinically relevant markers in heart failure prognosis.
- Echocardiography (Echo) Imaging: Echocardiographic studies provide key structural and functional cardiac measurements such as left ventricular ejection fraction (LVEF), wall motion abnormalities, and chamber dimensions. De-identified echo datasets will be sought via academic medical center collaborations. The EchoNet-Dynamic dataset (Stanford), containing over 10,000 apical-4-chamber echocardiogram videos with LVEF labels, will serve as a primary source for cardiac imaging data.
- Chest X-Ray Imaging: Chest radiographs will be incorporated to capture signs of pulmonary congestion, cardiomegaly, and pleural effusion — common findings in heart failure patients. The CheXpert dataset (Stanford) and NIH ChestX-ray14

dataset offer large-scale, labeled chest X-ray collections that can be used for pre-training image encoders and extracting imaging-derived features for downstream survival prediction.

4. Algorithms, Techniques, and Models

This project will adopt a comparative and progressive technical approach, beginning with established baselines and advancing to novel graph-based architectures.

Baseline Models (from Chicco & Jurman, 2020)

The baseline will replicate and extend the models evaluated in Chicco and Jurman (2020), including Linear Regression, Naive Bayes, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, Random Forests, Gradient Boosting, and ensemble methods. These models primarily target structured tabular clinical data and will serve as performance benchmarks.

Advanced Models: Graph Neural Networks and Graph Transformers

Beyond baselines, the core contribution of this project lies in applying Graph Neural Networks (GNNs) and Graph Transformer architectures to model the complex relational structure of patient data. Patients, clinical visits, diagnoses, medications, and lab values will be modeled as nodes and edges in a heterogeneous graph, enabling richer representation learning.

Data Pipeline

A unified data pipeline will be constructed to process all modalities:

- Structured clinical data (lab tests, vitals): Standard normalization, imputation, and tabular encoding.
- Unstructured doctor notes: Processed using clinical NLP models such as BioBERT or ClinicalBERT to extract embeddings, entities, and structured summaries.
- Wearable time-series data: Processed using temporal convolutional networks or LSTM encoders to generate fixed-length feature vectors.
- EHR history from EPIC: Transformed into graph-structured representations using FHIR-compatible ontologies.

Pre-existing implementations to be adopted include PyTorch Geometric (for GNN architectures), HuggingFace Transformers (for ClinicalBERT), and scikit-learn (for baseline models). Modifications will focus on adapting the Graph Transformer to handle heterogeneous node types and multi-modal edge features specific to clinical patient graphs.

5. Evaluation Strategy

Evaluation will closely follow the methodology of Chicco and Jurman (2020) to ensure comparability, while extending it to multimodal models.

Metrics

- Matthews Correlation Coefficient (MCC): Primary metric, particularly robust for imbalanced datasets.
- Area Under the ROC Curve (AUC-ROC): To measure discriminative power across thresholds.
- Precision, Recall, and F1-Score: To capture the trade-offs between sensitivity and specificity.
- Accuracy: For overall model correctness, reported alongside MCC to avoid misleading interpretations.

Plots and Visual Analysis

- ROC Curves: Plotted per model and overlaid for comparative visualization.
- Precision-Recall Curves: Especially informative given the class imbalance in survival outcomes.
- Confusion Matrices: Per model, to expose patterns in false positives and negatives.
- Feature Importance Plots: SHAP values or permutation importance to interpret which modalities and features drive predictions.
- Training/Validation Loss Curves: To diagnose overfitting and convergence behavior in deep models.

Qualitative Analysis

Case studies of individual patient predictions will be reviewed, particularly focusing on cases where the multimodal Graph Transformer significantly outperforms or underperforms relative to baseline classifiers. Clinical experts (where accessible) will be consulted to validate whether model-highlighted features align with established cardiology knowledge.

Appendix A: Architecture Diagrams

Figure 1 — End-to-End Data and ML Pipeline

The diagram below illustrates the full pipeline from raw multimodal data ingestion through modality-specific preprocessing, heterogeneous graph construction, parallel model training (baseline classifiers, GNN, and Graph Transformer), and final evaluation and explainability stages. Each data modality feeds into a dedicated encoder before being unified into the patient graph.

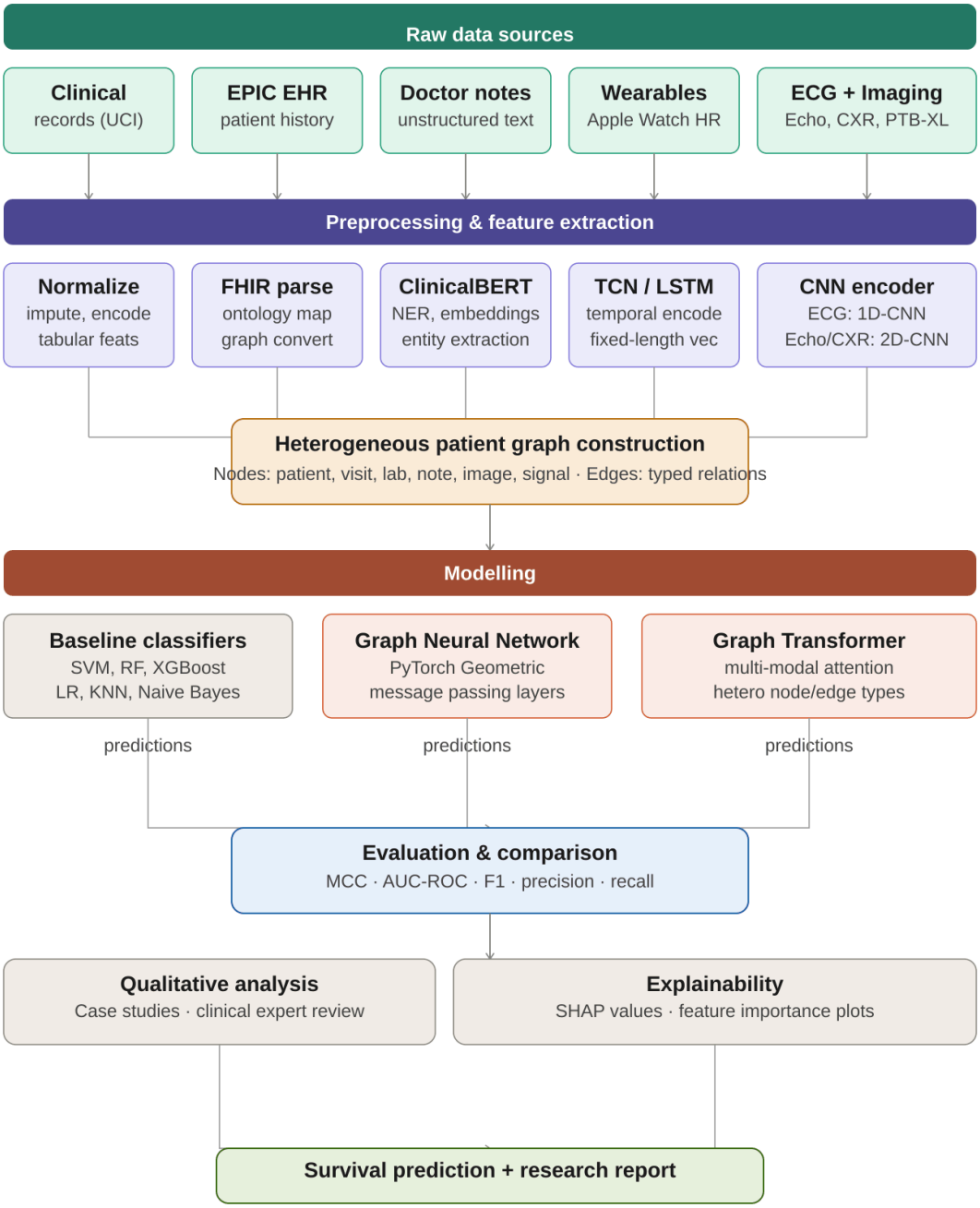

Figure 1. End-to-end data ingestion and ML pipeline for heart failure prediction.

Figure 2 — Graph Data Model: Nodes and Edges

The heterogeneous patient graph schema is shown below. The central Patient node connects to eight typed node classes — Clinical Visit, Lab Test, Diagnosis, Medication, Doctor Note, ECG Signal, Imaging Study, and Wearable Signal — via directed, labeled edges representing clinically meaningful relationships. Dashed edges indicate cross-node semantic relations (e.g., a Diagnosis informs a Medication). This graph

structure is the input representation learned over by the Graph Neural Network and Graph Transformer models.

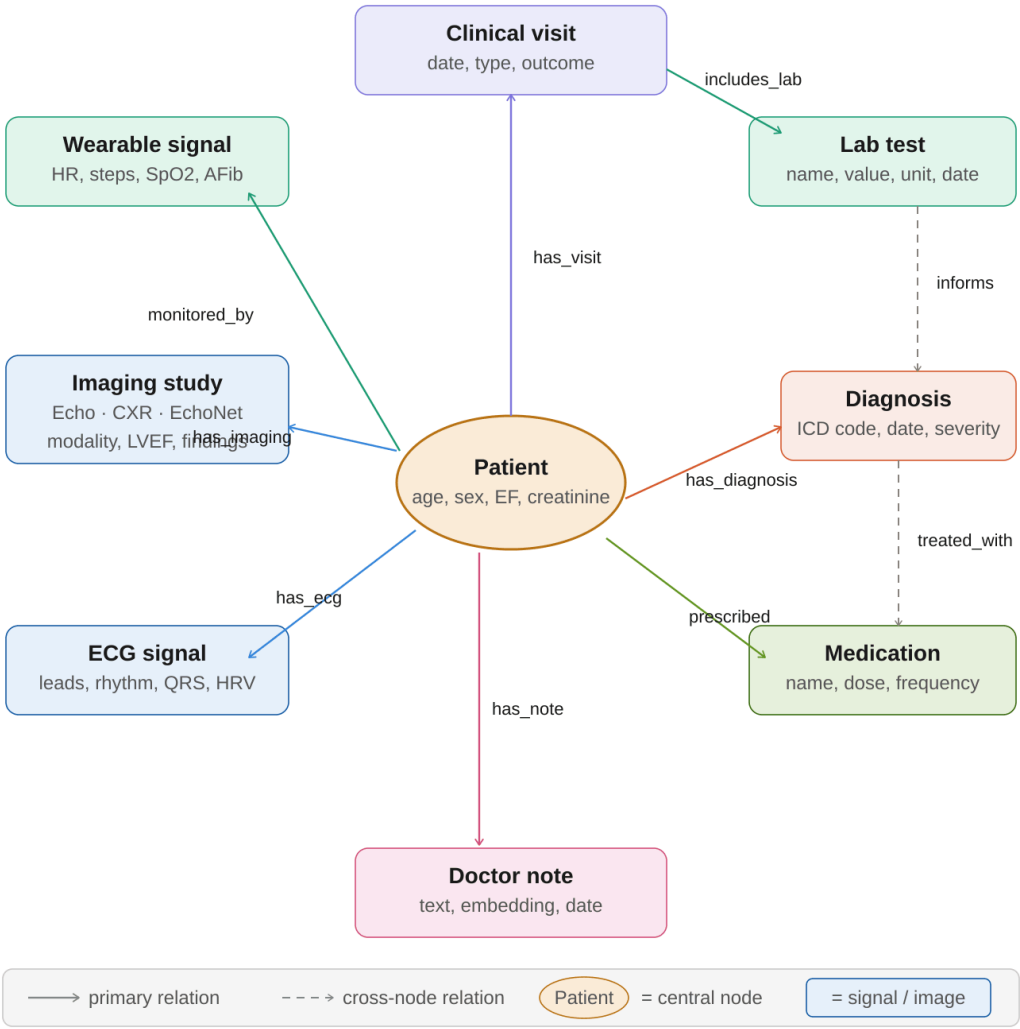

Figure 2. Heterogeneous graph data model showing node types and typed edge relations.

6. Post-Project Goals (Optional)

Upon completion, the goal is to prepare a research summary paper or poster suitable for submission to a student research symposium or workshop affiliated with a major ML/healthcare conference (e.g., ML4H, CHIL). Additionally, the data pipeline and model codebase will be open-sourced to enable reproducibility and extension by the broader research community.

7. References

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